

Q1: 24 Feb (Shift 2) - Single Correct

The calculated magnetic moments (spin only value) for species $[\text{FeCl}_4]^{2-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ and MnO_4^{2-} respectively are :

- (1) 5.92, 4.90 and 0 BM
- (2) 5.82, 0 and 0 BM
- (3) 4.90, 0 and 1.73 BM
- (4) 4.90, 0 and 2.83 BM

Q2: 25 Feb (Shift 1) - Single Correct

The hybridization and magnetic nature of $[\text{Mn}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$, respectively are:

- (1) d^2sp^3 and paramagnetic
- (2) $sp^3 d^2$ and paramagnetic
- (3) d^2sp^3 and diamagnetic
- (4) $sp^3 d^2$ and diamagnetic

Q3: 25 Feb (Shift 2) - Single Correct

Given below are two statements :

Statement I : The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH

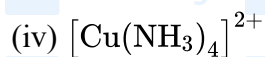
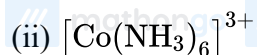
Statement II : The dimethyl glyoxime is a bidentate neutral ligand.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both statement I and statement II are true
- (2) Both statement I and statement II are false
- (3) Statement I is false but statement II is true
- (4) Statement I is true but statement II is false

Q4: 25 Feb (Shift 2) - Single Correct

If which of the following order the given complex ions are arranged correctly with respect to their decreasing spin only magnetic moment?



(1) (ii) > (i) > (iii) > (iv)

(2) (iii) > (iv) > (ii) > (i)

(3) (ii) > (iii) > (i) > (iv)

(4) (i) > (iii) > (iv) > (ii)

Q5: 25 Feb (Shift 2) - Numerical

The spin only magnetic moment of a divalent ion in aqueous solution (atomic number 29) is _____ BM.

Q6: 26 Feb (Shift 1) - Numerical

Number of bridging CO ligands in $[\text{Mn}_2(\text{CO})_{10}]$ is _____

Q7: 26 Feb (Shift 2) - Numerical

The number of stereoisomers possible for $[\text{Co}(\text{Ox})_2(\text{Br})(\text{NH}_3)]^{2-}$ is _____ [Ox = oxalate]

Answer Key**Q1 (3)****Q2 (1)****Q3 (4)****Q4 (4)****Q5 (2)****Q6 (0)****Q7 (3)**

Q1 (3)

$[\text{FeCl}_4]^{2-}$ Fe^{2+} 3 $d^6 \rightarrow$ 4 unpaired electron, as Cl^- in a weak field liquid.

$$\mu_{\text{spin}} = \sqrt{248} \text{M}$$

$$= 4.9 \text{BM}$$

$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ Co^{3+} 3 $d^6 \rightarrow$ for Co^{3+} with coordination no. 6 $\text{C}_2\text{O}_4^{2-}$ is strong field ligand & causes pairing & hence no. unpaired electron

$$\mu_{\text{spin}} = 0$$

$[\text{MnO}_4]^{2-}$ Mn^{+6} it has one unpaired electron.

$$\mu_{\text{spin}} = \sqrt{3} \text{BM}$$

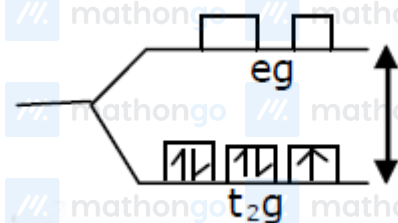
Q2 (1)

1. $(\text{Mn}(\text{CN})_6)^{4-}$

$\text{Mn}^{++} = 3 d^5$

$$\mu = \sqrt{3}$$

hybridization = d^2sp^3

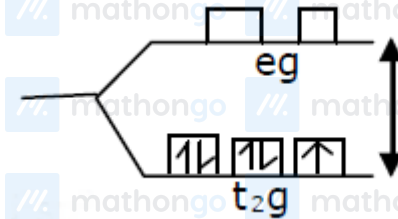


2. $[\text{Fe}(\text{CN})_6]^{3-}$

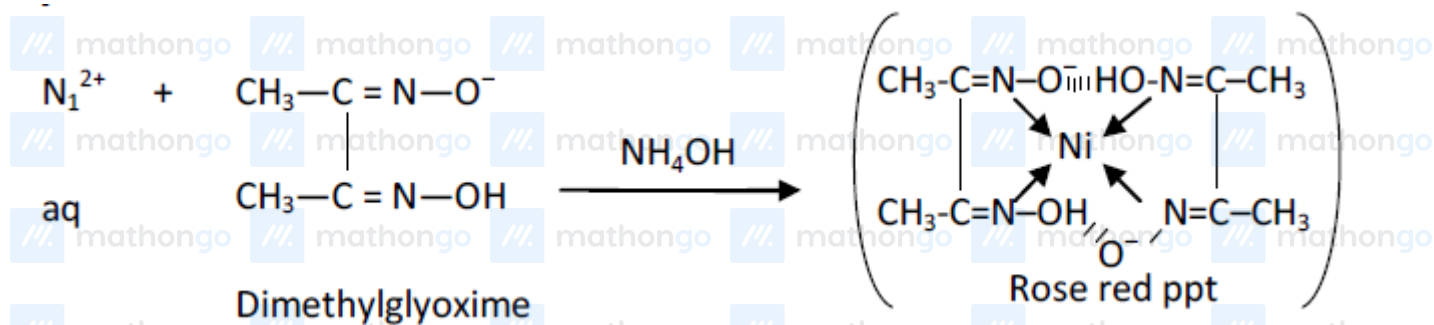
$\text{Fe}^{3+} = 3 d^5$

$$\mu = \sqrt{3}$$

Hybridization — d^2sp^3



Q3 (4)



Dimethyl glyoxime is a negative bidentate ligand.

Q4 (4)

$[\text{FeF}_6]^{3-}$ $\text{Fe}^{3+} 3d^5 \rightarrow 5$ -unpaired electrons as F^- is weak field ligand

$[\text{Co}(\text{NH}_3)_6]^{3+}$ $\text{Co}^{3+} 3d^6 \rightarrow$ No-unpaired electron as NH_3 is strong field ligand and causes pairing

$[\text{NiCl}_4]^{2-}$ $\text{Ni}^{2+} 3d^8 \rightarrow 2$ -unpaired electrons

$[\text{Cu}(\text{NH}_3)_4]^{2+}$ $\text{Cu}^{2+} 3d^9 \rightarrow 1$ -unpaired electrons

Q5 (2)

$_{29}\text{Cu}^{+2} \rightarrow [\text{Ar}]^{18} 3d^9$



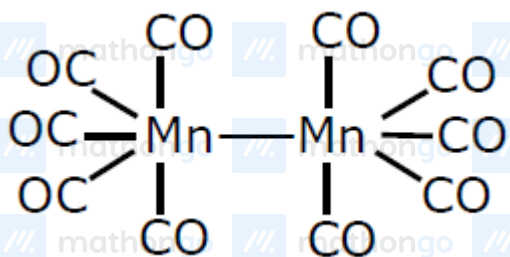
No. of unpaired $e^- = 1$ Magnetic moment $= \mu = \sqrt{n(n+2)}$

$$\mu = \sqrt{(1)(1+2)} = \sqrt{3} B.M$$

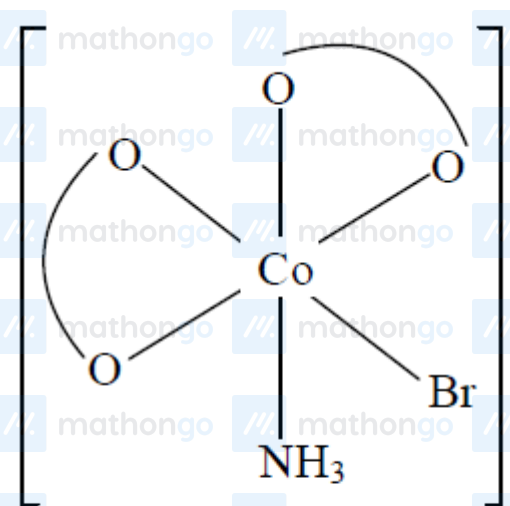
$= 1.73 \text{ Ans}$

Q6 (0)

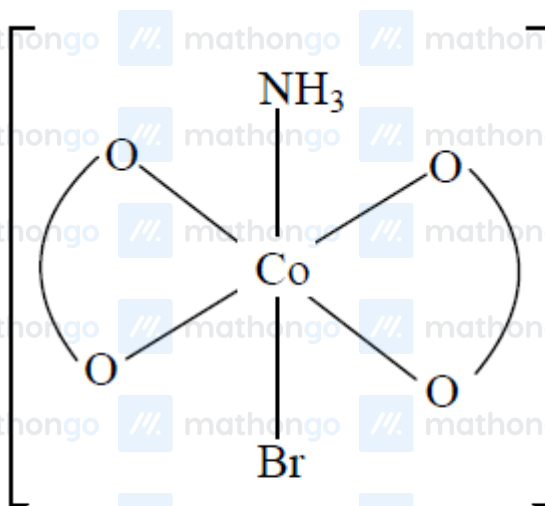
Fact



Q7 (3)



Optically active



Optically inactive



Mirror image

Total stereoisomer = 2(OI) + 1POE (pair of enantiomers) = 3